SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Questions

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

QUESTIONS: 5 **Max Marks50** **Duration :60 Minutes**

Annexure A

Scenario-Based Questions

Code of Conduct:

I (T. Varun Tej/192511217) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: T. Varun Tej

Annexure A

Scenario-Based Questions

1. **Dental X-Ray Enhancement and Cavity Region Extraction**

A dental clinic uses a digital radiography system to help dentists locate early-stage cavities (caries) in teeth. The X-ray images are dark and low in contrast, with quantum noise and overlapping tooth structures, so small cavity regions blend into the surrounding enamel and dentine.

Question: Apply suitable grey-level transformations (such as negative, log or power-law transformation) and histogram equalization to improve the visibility of the cavity regions. Select an appropriate spatial smoothing filter to suppress noise without blurring the cavity boundaries. Subsequently, apply thresholding or region growing to extract the suspected cavity region, and use edge detection to outline it. Explain the sequence of processing steps and justify the choice of each technique.

2. **Night-Time Vehicle Number Plate Region Extraction**

A traffic police department installs CCTV cameras at a toll gate to capture vehicles at night. The images suffer from very low brightness, headlight glare, sensor noise, and motion blur. The system must locate and extract the number-plate region so that the characters can be read in a later stage.

Question: Apply suitable grey-level transformations and histogram processing to enhance the dark regions while controlling glare. Use appropriate smoothing and sharpening filters (for example, median filtering followed by Laplacian or unsharp masking) to reduce noise and restore character edges. Apply edge detection, line detection and edge linking to locate the rectangular plate boundary, and use thresholding to separate the characters from the plate background. Present the complete processing pipeline and explain how each step contributes to extracting the number-plate region.

3. **Textile Fabric Defect Detection Using Frequency-Domain Processing**

A textile mill inspects woven fabric rolls using a line-scan camera. The fabric has a regular, repeating weave pattern, and defects such as holes, broken threads, oil stains and knots disturb this pattern. The captured images also contain periodic noise from the machine lighting and uneven illumination across the width of the roll.

Question: Apply suitable frequency-domain filtering techniques (low-pass, high-pass, or notch/band-reject filtering) to suppress the periodic noise and the regular weave texture, so that the defects stand out. Use histogram processing to correct the uneven illumination. Subsequently, apply thresholding and boundary detection to segment and outline the defective regions. Explain the order of operations and justify why frequency-domain processing is preferred over spatial-domain processing in this scenario.

4. **Blood Cell Segmentation in Microscopic Images**

A pathology laboratory uses an automated microscope to capture images of stained blood smears for counting red and white blood cells. The images show uneven staining, poor contrast between cells and background, touching or overlapping cells, and noise from dust particles on the slide.

Question: Apply suitable contrast enhancement and histogram processing techniques to improve the separation between cells and background. Use an appropriate spatial filter to remove small noise particles while preserving cell boundaries. Apply global or adaptive thresholding to separate cells from the background, and use region-based segmentation (region growing or region splitting and merging) to separate touching cells. Finally, apply edge detection and boundary detection to outline individual cells. Explain the processing sequence and the role of each technique in obtaining accurate cell regions.

5. **Text Extraction from Degraded Historical Documents**

A state archive is digitising old handwritten manuscripts. The scanned pages contain yellowed paper, ink bleed-through from the reverse side, stains, faded strokes and uneven lighting across the page. The archive requires clean extraction of text regions before handwriting recognition.

Question: Apply suitable grey-level transformations and histogram processing to improve the contrast between faded ink and the background. Select appropriate smoothing and sharpening filters to remove stains and noise while strengthening the pen strokes. Compare global thresholding (such as Otsu's method) with local (adaptive) thresholding for this scenario and apply the more suitable one to separate text from the background. Use edge detection and boundary detection to identify text lines and word regions. Explain the sequence of operations and justify each decision.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

1. Dental X-Ray: Main Problems and Enhancement

Scenario 1

Problems: Dark, low-contrast X-rays; quantum noise; overlapping tooth structures; small cavities blend with enamel and dentine.

Grey-level enhancement: Use power-law (gamma < 1) to brighten dark regions. Apply histogram equalization or CLAHE to improve local contrast.

Negative transformation: Can be tested if cavities are easier to inspect when intensity relationships are inverted; compare visually with the original.

Noise removal: Use a small median filter for impulse-like noise, or a mild Gaussian filter for random noise. Keep the kernel small to protect cavity edges.

1. Dental X-Ray: Segmentation Pipeline

Recommended order

1. Preprocess: Convert to grayscale if needed and normalize intensity.

2. Enhance: Apply gamma correction followed by CLAHE/histogram equalization to reveal subtle dark regions.

3. Smooth: Apply mild median or Gaussian smoothing to reduce noise without erasing small cavity boundaries.

4. Segment: Use adaptive thresholding when brightness varies, or region growing from a suspected cavity seed with similar intensity.

5. Outline: Use Canny edge detection and boundary tracing to outline the suspected cavity.

Validation: Treat the output as a suspected region; a dentist must confirm the diagnosis.



1. Dental X-Ray: Why These Methods?

Justification

Gamma correction: Brightens dark pixels and makes low-intensity details more visible.

CLAHE / histogram equalization: Improves contrast so small intensity differences become easier to see; CLAHE limits excessive local amplification.

Small smoothing filter: Reduces noise while limiting loss of fine cavity edges.

Adaptive threshold / region growing: Separates a candidate cavity based on intensity and local similarity; region growing can follow connected areas.

Edge detection: Provides a clear outline for review and measurement.

2. Night-Time Number Plate: Enhancement

Scenario 2

Problems: Very low brightness, headlight glare, sensor noise and motion blur.

Brightness and contrast: Use gamma correction (gamma < 1) for dark areas and CLAHE for local contrast.

Glare control: Avoid globally brightening the entire image; use local contrast enhancement and, where possible, mask saturated headlight regions.

Noise reduction: Apply a median filter to suppress isolated noise while preserving edges.

Sharpening: Use unsharp masking or a mild Laplacian-based sharpening step after denoising. Sharpening cannot fully recover severe motion blur.

2. Number Plate: Locate and Extract

Recommended order

1. Detect edges: Apply Canny edge detection to identify strong plate and character boundaries.
2. Find straight lines: Use Hough line detection to find likely horizontal and vertical plate edges.
3. Link edges: Use morphological closing/edge linking to connect broken boundary segments.
4. Select plate: Choose a rectangular candidate with plausible aspect ratio and strong edge density.
5. Crop and threshold: Crop the plate, convert to grayscale if needed, then use adaptive thresholding to separate characters from the plate background.
6. Output: Return the cropped plate and binary character image for later OCR.



2. Number Plate: Justification

Justification

Median filter: Removes small noise without excessively blurring character edges.

Unsharp masking / Laplacian: Enhances edge transitions, making character strokes more distinct.

Canny + Hough: Canny provides edge pixels; Hough detection identifies straight lines that form a rectangular plate.

Edge linking: Connects gaps caused by noise or weak contrast.

Adaptive thresholding: Handles uneven lighting across the plate better than one global threshold.

3. Textile Defects: Frequency-Domain Processing

Scenario 3

Problems: Regular repeating weave, defects such as holes/broken threads/oil stains/knots, periodic lighting noise and uneven illumination.

Correct illumination: Use background estimation and normalization, or local histogram processing/CLAHE, to reduce brightness variation across the fabric.

Transform: Apply the Fourier Transform to represent image patterns by spatial frequency.

Periodic noise: Use notch-reject filters at identified noise-frequency peaks.

Weave texture: Use carefully selected band-reject or high-pass/low-pass filtering depending on whether the weave or defect detail occupies the target frequency range.

3. Textile Defects: Segment and Outline

Recommended order

1. Illumination correction: Normalize the background or apply local contrast enhancement.
2. Frequency analysis: Compute the Fourier spectrum and identify peaks caused by periodic lighting or weave texture.
3. Filter: Apply notch filtering for periodic noise; use band-reject filtering only when the unwanted weave frequencies can be separated from defect frequencies.
4. Reconstruct: Apply the inverse Fourier Transform to return to the image domain.
5. Segment: Threshold the filtered image to isolate unusual regions; choose global or local thresholding based on residual illumination variation.
6. Outline: Use boundary detection/contours to mark defect regions.



3. Why Frequency-Domain Filtering?

Justification

Periodic patterns: Repeated weave and periodic lighting produce concentrated frequency components that can be identified in the Fourier spectrum.

Targeted removal: Notch filters can suppress specific periodic-noise frequencies rather than blurring the entire image.

Important caution: The weave and defects may overlap in frequency. Over-filtering can remove real broken threads or small holes.

Spatial methods still help: Thresholding and boundary detection are applied after filtering to locate and outline the defects.

Best practice: Inspect the spectrum and compare filtered images before selecting filter frequencies.

4. Blood Cell Segmentation: Enhancement

Scenario 4

Problems: Uneven staining, low contrast between cells and background, touching/overlapping cells and dust particles.

Contrast enhancement: Use histogram equalization or CLAHE to improve visibility of cell regions while limiting over-enhancement.

Noise removal: Use a small median filter to remove isolated dust-like noise; avoid strong smoothing that distorts cell boundaries.

Thresholding: Use adaptive thresholding when staining/illumination varies across the image; global thresholding may work if the background is uniform.

4. Blood Cells: Separate and Outline

Recommended order

1. Preprocess: Convert to a suitable grayscale or color channel that separates stained cells from the background.

2. Enhance and denoise: Apply CLAHE and a mild median filter.

3. Initial segmentation: Apply adaptive thresholding to create a foreground mask for cells.

4. Separate touching cells: Use region growing based on similar intensity/color, or region splitting and merging. Distance transform and watershed can also help separate touching objects.

5. Boundary detection: Detect edges and trace contours around each segmented cell.

6. Count and check: Count individual regions and inspect merged/split errors before reporting results.



4. Blood Cells: Why These Methods?

Justification

CLAHE: Improves local cell-to-background contrast under uneven staining.

Median filter: Removes small isolated particles while preserving boundaries better than strong averaging.

Adaptive threshold: Uses local image neighborhoods, so it is more robust to non-uniform brightness.

Region-based segmentation: Groups connected pixels with similar properties; splitting/merging helps handle touching cells.

Boundary detection: Provides outlines needed for cell counting and shape analysis.

5. Historical Documents: Enhancement

Scenario 5

Problems: Yellowed paper, bleed-through, stains, faded ink strokes and uneven illumination.

Grey-level transformation: Use contrast stretching or a suitable power-law transformation to make faded strokes more visible.

Histogram processing: Use histogram equalization or CLAHE to improve ink-paper contrast; CLAHE is often safer when lighting varies locally.

Noise and stain removal: Use mild median filtering for small specks and background estimation/normalization for slow illumination changes.

Sharpening: Apply mild unsharp masking to strengthen pen strokes, but avoid amplifying bleed-through and stain edges.

5. Global vs Adaptive Thresholding

Thresholding comparison

Global / Otsu: Chooses one threshold for the whole page. It is simple and effective when foreground and background intensities are well separated and illumination is fairly uniform.

Adaptive thresholding: Calculates a threshold from local neighborhoods. It handles uneven lighting and locally faded ink better.

Best choice here: Adaptive thresholding is generally more suitable because the page has uneven illumination, stains and faded strokes.

Caution: Choose neighborhood size carefully: too small may preserve stains/noise; too large may fail to follow local lighting changes.

5. Text Regions: Final Pipeline

Recommended order

1. Normalize background: Estimate and correct slow illumination variation across the page.
2. Enhance contrast: Apply contrast stretching or CLAHE to distinguish ink from paper.
3. Denoise and sharpen: Use mild median filtering, then gentle sharpening if required.
4. Binarize: Apply adaptive thresholding to separate text strokes from the background.
5. Find text lines: Use connected components, horizontal projection or morphology to group characters into lines.
6. Boundaries: Trace bounding boxes/contours around text lines and word regions before handwriting recognition.

